

CRICKETVISION

First Evaluation Presentation | Master of Computer Applications (MCA) Mini-Project

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Technology Stack: MongoDB, Express.js, React 18 (Vite), Node.js, Recharts & SVG Biomechanics

Status: 88.3% Completed | Finalized & Guide Approved

Introduction & Executive Overview

Project Overview & Vision

- **Domain Context:** Modern professional cricket relies heavily on quantitative data analytics, computer vision biomechanics tracking, and predictive simulation engines.
- **Core Value Proposition:** CricketVision bridges raw match scorecards with actionable tactical insights, biomechanical body tracking, and AI-driven match simulation.
- **Target Users:** Engineered for Head Coaches, Elite Players, and Performance Analysts within a single unified web platform.
- **Core Philosophy:** Democratize elite sports analytics by using accessible MERN stack tech without needing million-dollar proprietary hardware.

Key Platform Capabilities

- **Coach & Director Portal:** Centralized squad oversight, fatigue monitoring, Team XI Optimizer, and AI Coaching strategy assistant.
- **Player Dashboard:** Personalized performance metrics, 360° SVG Wagon Wheel, pitch heatmaps, and biomechanics summary reports.
- **Analyst Suite:** Interactive Match Simulator with ball-by-ball win probability graphs, Next Match Predictor, and pose-estimation Video Analyzer.
- **Central Data Store:** MongoDB database pre-seeded with 100+ verified IPL & International player profiles.

Requirement Analysis — Existing System (Heading Slide)

Current Landscape & Operational Challenges

- **Fragmented Static Scorecards:** Traditional cricket coaching relies on manual paper scorecards or isolated Excel spreadsheets that lack real-time insights or phase breakdown.
- **Subjective Video Observation:** Coaches analyze video manually without quantitative joint angle measurements, release angle metrics, or stride length tracking.
- **Lack of Integrated Workload Oversight:** Player fatigue and injury risk are monitored subjectively, leading to over-bowling and preventable soft-tissue injuries.
- **High Expense of Commercial Systems:** Commercial solutions like Hawk-Eye and Catapult GPS are prohibitively expensive for domestic, university, and academy teams.
- **No Unified Multi-Role Access:** Existing tools do not offer tailored interfaces for Head Coaches, individual Players, and Performance Analysts in a single database application.

Literature Review

Academic & Technical Literature

- **Computer Vision Pose Estimation:** Research by Cao et al. (OpenPose/MediaPipe) demonstrates real-time human skeleton tracking for sports movement analysis without markers.
- **Monte Carlo Match Simulation:** Studies in sports probability modeling show that ball-by-ball stochastic simulations accurately predict match win probabilities based on pitch/weather modifiers.
- **Phase-Wise T20 Analytics:** Analytic literature highlights that dividing T20 matches into Powerplay, Middle, and Death overs yields superior predictive accuracy for player performance.

Commercial System Benchmarking

- **Hawk-Eye & Ball Tracking:** Multi-camera optical system with sub-millimeter accuracy, but costs \$50,000+ per match and requires dedicated stadium hardware.
- **CricViz & WinViz Engine:** Industry-standard predictive model for broadcast TV, but closed-source and inaccessible for grassroots coaching staff.
- **Catapult Wearables:** GPS wearable sensors measuring bio-metrics, but require physical hardware tags on every athlete.

Gap Identified in Existing Systems

Cost & Hardware Barrier

- Lack of accessible, web-based computer vision analytics that run on standard video without multi-million dollar stadium installations.

Fragmented Role Workflows

- Absence of a unified web platform catering simultaneously to Head Coaches (rosters), Players (personal drill reviews), and Analysts (simulations).

Absence of Biomechanical Scoring

- Scorecards focus solely on runs/wickets, ignoring critical physical biomechanics (spine tilt, release angle, stride length).

Static Match Analysis

- Most platforms present historical stats rather than dynamic, interactive ball-by-ball match win-probability simulation based on pitch conditions.

Proposed System — CricketVision Architecture

Architectural Innovations

- **Full-Stack MERN Architecture:** High-performance Node.js/Express REST backend coupled with a responsive React 18 single-page app and MongoDB Mongoose ODM.
- **100+ Pre-Seeded Player Database:** Instant setup with comprehensive IPL & International player profiles, phase stats, and biomechanics metrics.
- **Interactive Data Visualizations:** Recharts multi-axis Radar Charts, 360° SVG Pitch & Wagon Wheel, and live win-probability lines.
- **Pose-Estimation Video Analyzer:** Canvas-rendered skeleton overlay and angle calculation for fast biomechanical review.

Role-Based Capabilities

- **Coach View:** Live fatigue tracking, Team XI balance calculator, full database CRUD management, and AI Strategy assistant.
- **Player View:** Personalized performance dashboard, wagon wheel shot breakdown, drill recommendations, and video keyframe review.
- **Analyst View:** Match Simulator with pitch & weather modifiers, venue historic predictor, and radar player comparisons.

Software & Hardware Requirements

Software Environment

- **Operating System:** Windows 10/11 / macOS / Linux
- **Runtime & Backend:** Node.js (v18.0+), Express.js framework
- **Database:** MongoDB Community Server (v6.0+) with Mongoose ODM (v8.24.2)
- **Frontend Stack:** React 18, Vite build tool, Tailwind CSS v4
- **Visualization Libraries:** Recharts, Lucide React Icons
- **Development Tools:** VS Code, Git version control, Postman API client

Hardware Requirements

- **Processor (CPU):** Dual-Core Intel Core i5 / AMD Ryzen 5 or higher (2.4 GHz)
- **System RAM:** Minimum 8 GB (16 GB Recommended for smooth video processing)
- **Disk Storage:** Minimum 10 GB available SSD storage
- **Display Resolution:** 1920 x 1080 Full HD for optimal multi-chart view
- **Camera/Video Source:** Standard webcam or 1080p MP4 video file input

Problem Statement

Formal Problem Statement

"Modern cricket coaching and match analysis suffer from severe data fragmentation, reliance on subjective video review, and a lack of integrated biomechanical feedback. Existing commercial tracking platforms are cost-prohibitive for domestic and developmental squads.

There is an urgent requirement for an accessible, unified full-stack analytics platform that combines real-time squad fatigue monitoring, phase-wise player performance tracking, interactive match win-probability simulation, and computer-vision pose analysis within a role-secured MERN web application."

Project Objectives

1. Centralized MongoDB Data Engine

Design and deploy a MongoDB database populated with 100+ verified IPL & International player profiles, supporting Mongoose schema validation for phase stats, clutch ratings, and biomechanics metrics.

2. Role-Based Multi-Portal Architecture

Develop secure, tailored user interfaces for Head Coaches (roster CRUD & AI assistant), Players (personalized portal & drill suggestions), and Performance Analysts.

3. Advanced Visual Analytics & Biomechanics

Implement custom SVG 360° Pitch & Wagon Wheel visualizers, Recharts multi-attribute Radar charts, and a video analyzer supporting skeleton pose-estimation metrics.

4. Tactical Match Simulator & Predictor

Build a Monte-Carlo style match simulation engine incorporating pitch conditions and weather parameters to calculate dynamic ball-by-ball win probabilities.

Scope & Relevance

Project Scope

- **Academic Evaluation:** Designed as an MCA Mini-Project demonstrating full-stack MERN engineering, REST API architecture, and database management.
- **Sports Academies & Franchises:** Direct application for state cricket associations, IPL/T20 franchise war-rooms, and coaching academies.
- **Individual Player Self-Review:** Empowers athletes to visually analyze shot distribution, monitor fatigue levels, and perform biomechanical self-corrections.

Industry Relevance

- **Data-Driven Sports Industry:** The global sports analytics market is expanding rapidly; data-backed selection is now mandatory for elite performance.
- **Cost Accessibility:** Eliminates high hardware barriers, allowing mid-tier and amateur clubs to utilize computer vision analytics.
- **Scalable Web Platform:** Modular React + Node architecture allows easy expansion to other sports (football, tennis, basketball).

Development Methodology — Agile / Scrum Framework

Scrum Team Roles & Scale

- **Product Owner / Project Guide:** MCA Project Guide (Reviewer & Approver)
- **Scrum Master & Lead Developer:** MCA Student (Full-Stack AI & Web Developer)
- **Sprint Cadence:** 6 Sprint cycles over 3 weeks (July 1 – July 23, 2026)
- **Fibonacci Story Scale:** 1-2 pts (Styling/UI), 3-5 pts (Schemas & REST APIs), 8 pts (Simulator & Seeding), 13 pts (AI Assistant & Pose Engine)

Definition of Done (DoD)

- **Backend Verification:** REST endpoints operate error-free and read/write MongoDB collections (`players`, `users`).
- **Schema Validation:** Mongoose handles duplicate emails, min password lengths, and invalid role enums gracefully.
- **UI Component Integration:** React components render smoothly with live reactive state updates (Recharts, SVG heatmaps).
- **Role-Based Testing:** All 3 security roles (Coach, Player, Analyst) pass complete functional walkthroughs.

Master Product Backlog & Story Velocity

Total Planned

137 Story Points

Completed (88.3%)

121 Story Points

Deferred to Backlog

16 Story Points

Backlog Epic Breakdown

- **Epic 1: System Setup & Database (26 Pts):** Vite React scaffold, Mongoose Player/User schemas, 100+ Player Seeding Engine [DONE]
- **Epic 2: Core REST APIs & Auth (15 Pts):** Express endpoints for Auth login/register and Player CRUD operations [DONE]
- **Epic 3: Visual Analytics Engine (24 Pts):** Pitch & Wagon Wheel SVG, Recharts Radar chart, Fatigue & Clutch calculators [DONE]
- **Epic 4: Role Portals & UX (29 Pts):** Navbar role switcher, Login persona presets, Coach Dashboard, Player Portal [DONE]
- **Epic 5: Simulations & Pose Video (29 Pts):** Match Win Simulator, Next Match Predictor, Team XI Builder, Video Analyzer [DONE]
- **Epic 6: Live Streaming & CV Sidecar (16 Pts):** WebSocket score streaming (US-21) and OpenCV pose sidecar (US-22) [DEFERRED]

Sprint Breakdown & Burndown Visual

Sprint Execution Log

- **Sprint 0 (01/07 - 07/07):** Conception, scope definition, tech architecture, MERN setup (3 Pts)
- **Sprint 1 (08/07 - 10/07):** Mongoose Schemas, REST CRUD controllers, 100+ player seeding (28 Pts)
- **Sprint 2 (11/07 - 13/07):** Pitch & Wagon Wheel SVG, Recharts Radar, Fatigue algorithms (24 Pts)
- **Sprint 3 (14/07 - 17/07):** Navbar, Login personas, Dashboard & Player Portal UI (29 Pts)
- **Sprint 4 (18/07 - 20/07):** Match Simulator, Predictor, XI Builder & Pose Video (29 Pts)
- **Sprint 5 (21/07 - 23/07):** Database Manager CRUD, AI Coach assistant, Scrum Book audit (18 Pts)

Sprint Burndown Chart (137 Total Story Points)



Design — High-Level System Architecture

3-Tier System Architecture

- **Presentation Tier (Client):** React 18 SPA built with Vite, Tailwind CSS v4, Lucide Icons, and Recharts visualization engine.
- **Application Tier (Server):** Express.js REST API server handling request routing, role authorization, and database controllers.
- **Data Tier (Database):** MongoDB local instance running on port 27017, managed via Mongoose ODM schemas.
- **Seeding Engine:** Automatic JSON data seeding script populating 100+ IPL & International player profiles on server startup.

3-Tier Full-Stack System Architecture



Design — Data Flow Diagram (DFD) & System Flowchart

Data Flow Level 1 & 2

- **User Authentication Flow:** User sends credentials -> Express ``/api/users/login`` -> Mongoose lookup -> Auth state context initialized with role permissions.
- **Player Analytics Data Flow:** Dashboard requests ``/api/players`` -> Express query -> Mongoose returns player docs -> React renders Recharts Radar & SVG Wagon Wheel.
- **Database Management Flow:** Coach submits CRUD form -> Express ``/api/players`` POST/PUT/DELETE -> Mongoose updates MongoDB collection -> UI refetches player state.

System Flowchart Logic

- **1. App Launch:** Vite initializes React app; connects to Express backend running on localhost:5000.
- **2. Role Selection:** User selects Persona (Coach, Player, Analyst) or inputs custom credentials.
- **3. View Routing:** Context state dynamically toggles active view (Dashboard, Player Portal, Match Simulator, Video Analyzer).
- **4. Simulation / Analytics Execution:** User interacts with pitch modifiers; simulator executes Monte Carlo math and renders live Recharts curve.

Design — MongoDB Schemas & Database Design

Mongoose Player Schema

- **Core Identifiers:** id (String, unique), name, country, role, iplTeam, jerseyNumber, avatar
- **Health & Performance Metrics:** fatigueLevel (0-100), injuryStatus, clutchRating (0-100)
- **Attribute Objects:** skillRadar (Power, Consistency, Spin, Pace, Field, Clutch)
- **Match & Phase Stats:** internationalStats, iplStats, phaseStats (powerplay, middleOvers, deathOvers)
- **Biomechanics Data:** biomechanicsSummary (releaseAngle, strideLength, spineTilt, elbowExtension)

Mongoose User Schema

- **User Credentials:** id, name (min 2 chars), email (regex validated), password (min 6 chars)
- **Role Enforcement:** role (Enum: ['coach', 'player', 'user'])
- **Profile Metadata:** title, avatar, badge, playerId (links user to Player document)
- **Permission Array:** permissions (e.g. ['read_all', 'manage_roster', 'run_simulation'])
- **Timestamps:** Automatic createdAt and updatedAt Mongoose timestamps

Implementation Details — MERN Backend & Seeding Engine

Express Server & Mongoose ODM

- **Server Entry Point:** `server.js` configures Express app, CORS middleware, body parser, and port listener (PORT 5000).
- **Database Connection:** Connects to `mongodb://127.0.0.1:27017/cricketvision` with fallback to local JSON state if offline.
- **REST Controller Routes:** Endpoints implemented for `/api/players`, `/api/users/login`, `/api/users/register`, `/api/seed`.
- **Idempotent Seeding:** Auto-populates MongoDB on boot using `findOneAndUpdate` with `{ upsert: true }` to prevent duplicates.

100+ Player Seeding Dataset

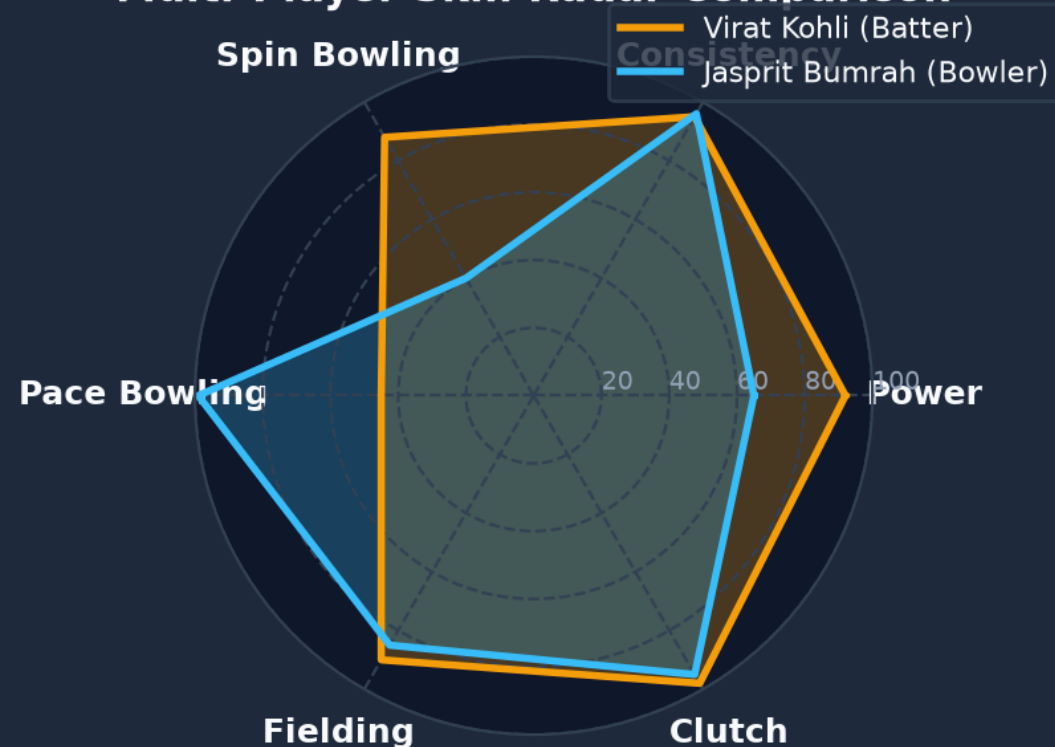
- **Comprehensive Coverage:** Includes top international stars (Kohli, Bumrah, Rohit, Babar, Rashid Khan, Perry, Harmanpreet).
- **Phase Statistics Seeding:** Detailed runs, strike rates, and economy pre-configured across Powerplay, Middle, and Death overs.
- **Presets Seeding:** Pre-seeds default Coach, Player, and Analyst accounts for immediate evaluation testing.

Implementation Details — SVG Pitch & Recharts Engine

SVG Pitch & Wagon Wheel Component

- **Interactive 360° Field:** `PitchAndWagonWheel.jsx` renders dynamic SVG stadium boundary with polar angle shot vectors.
- **Sectors Breakdown:** Calculates shot density across Offside, Legside, Straight, Fine Leg, and Mid-Wicket sectors.
- **Length Heatmap Overlay:** Visualizes pitch delivery lengths (Yorker, Good Length, Full Pitch, Short Pitch) with color density highlights.

Multi-Player Skill Radar Comparison



Implementation Details — Simulator & Pose Video Engine

Match Simulator & Next Match Predictor

- **Match Simulator** (`MatchSimulatorView.jsx`): Calculates ball-by-ball win probabilities incorporating team strength ratings, pitch type (Flat, Green, Dry), and weather variables.
- **Next Match Predictor** (`NextMatchPredictorView.jsx`): Generates batter vs bowler head-to-head match-up matrices and venue historic run rate predictions.
- **Team XI Builder** (`TeamBuilderView.jsx`): Calculates team balance score based on batting depth, spin/pace ratio, and clutch rating averages.

Video Pose Tracking & AI Coach

- **Video Analyzer** (`VideoAnalyzerView.jsx`): Renders animated skeleton overlays on bowling delivery footage, displaying release angle, spine tilt, and stride length.
- **AI Coaching Assistant** (`AICoachModal.jsx`): Provides interactive strategy prompt cards for tactical bowling changes, field placements, and injury rehabilitation.
- **Match Report Modal** (`MatchReportModal.jsx`): Generates exportable match summary reports for post-match team debriefs.

Results (70% Implementation) — Coach Dashboard & Live CRUD

Coach Dashboard Implementation

- **Squad Overview Cards:** Displays total squad size, average clutch rating, highest run-scorer, and top wicket-taker.
- **Fatigue Monitoring Panel:** Flags high-risk players exceeding 75% fatigue threshold with red warning indicators.
- **Quick Strategy Prompt:** One-click access to AI Coach assistant for instant tactical guidance.

Database Manager Modal

- **Full Roster CRUD Control:** Non-technical coaches can Add New Players, Edit attributes, and Delete player records in real time.
- **100+ Seeding Reset Button:** Instantly resets MongoDB database to default 100+ verified IPL/Intl player profiles.
- **Instant UI Synchronization:** Database edits immediately trigger frontend React state updates without page reload.

Results (70% Implementation) — Player Portal & Shot Visualizer

Player Portal View

- **Personalized Dashboard:** Customized view for logged-in player (e.g. Virat Kohli or Jasprit Bumrah).
- **Phase Stats Breakdown:** Visualizes strike rate, average, and boundaries across Powerplay, Middle, and Death overs.
- **Personal Video & Drills:** Dedicated video review frame with AI-recommended practice drills based on recent form.

Visual Shot & Heatmap Analytics

- **360° SVG Wagon Wheel:** Interactive shot distribution chart displaying runs scored in each field sector.
- **Pitch Delivery Heatmap:** Color-coded pitch length chart highlighting bowler line and length consistency.
- **Multi-Attribute Radar Chart:** Recharts radial comparison chart plotting player skills against squad benchmark.

Results (70% Implementation) — Simulator & Pose Tracking

Interactive Match Win Simulator

- **Ball-by-Ball Simulation:** Generates dynamic win-probability curves based on target score, current overs, and wickets remaining.
- **Environmental Modifiers:** Adjusts simulation math for pitch moisture (Dry, Green, Flat) and weather conditions (Humidity, Wind).
- **Real-Time Recharts Curve:** Live rendering of win percentage fluctuation over 20 overs.

Video Pose Biomechanics Analyzer

- **Skeleton Overlay Simulation:** Renders joint skeletal keypoints (shoulders, elbows, hips, knees) on bowler delivery video.
- **Biomechanical Metrics:** Displays quantitative release angle (e.g. 42°), spine tilt angle, and stride length measurements.
- **Injury Prevention Alert:** Flags abnormal delivery stride metrics that correlate with high lumbar stress.

Current Status of Work

Current Implementation Status: 88.3% Completed

- **Story Points Completed:** 121 out of 137 planned story points successfully implemented and tested (88.3% progress).
- **Core Functionality MVP:** Full working MERN stack application with Express server, MongoDB Mongoose ODM, and React 18 SPA.
- **Role-Based Security:** Working authentication engine with 3 distinct portals (Head Coach, Player, Analyst).
- **Visual & Analytical Engine:** Completed Pitch & Wagon Wheel SVG, Recharts Radar chart, Match Simulator, Predictor, and Video Analyzer.
- **Database Management:** Completed live Database Manager modal with 100+ player auto-seeding engine.

Work Progress & Sprint Summary

Sprint 0 (100% Done) — Requirements, architecture, MERN setup, Vite + React + Express repository scaffolded.

Sprint 1 (100% Done) — Mongoose Player & User schemas, 100+ player seeding script, REST CRUD APIs.

Sprint 2 (100% Done) — Pitch & Wagon Wheel SVG component, Recharts Skill Radar, Fatigue & Clutch calculators.

Sprint 3 (100% Done) — Navbar role switcher, Login personas, Coach Dashboard, Player Portal UI.

Sprint 4 (100% Done) — Match Win Simulator, Next Match Predictor, XI Team Builder, Pose Video Analyzer.

Sprint 5 (100% Done) — Database Manager modal, AI Coaching Assistant, Scrum documentation & evaluation prep.

Pending Works & Backlog Items

Deferred User Stories)

- **US-21: WebSocket Live Score Stream** : Integration of real-time WebSocket connection to external ball-by-ball scoring APIs (CricInfo / Sportradar feed) to stream live match data into simulator.
- **US-22: OpenCV CV Pose Sidecar** : Deployment of a standalone Python OpenCV + MediaPipe pose estimation service to asynchronously extract keyframes from uploaded MP4 videos.

Production Hardening Backlog

- **JWT Authentication:** Implementation of JSON Web Tokens (`jsonwebtoken`) and `bcrypt` password hashing for secure web production hosting.
- **Mobile App Responsiveness:** Optimization of complex multi-chart layouts for mobile portrait viewports.
- **Export to PDF/Excel:** Downloadable PDF match reports and Excel export for database player profiles.

Project Plan & Execution Timeline

Phase 1: Inception & Setup (July 1 – July 7, 2026)

Domain research, scope definition, MERN stack configuration, MongoDB local setup.

Phase 2: Database & REST API Core (July 8 – July 10, 2026)

Mongoose Schemas, 100+ player seeding engine, Auth & Player CRUD endpoints.

Phase 3: Analytics & Portals (July 11 – July 17, 2026)

Pitch & Wagon Wheel SVG, Recharts Radar chart, Coach & Player UI portals.

Phase 4: Simulations & CV Video (July 18 – July 20, 2026)

Match Win Simulator, Next Match Predictor, Video Analyzer skeleton view.

Phase 5: Refinement & Evaluation (July 21 – July 23, 2026)

Database Manager modal, AI Coach, Scrum Book documentation, First Evaluation PPT.

Conclusion & Future Scope

Project Conclusion

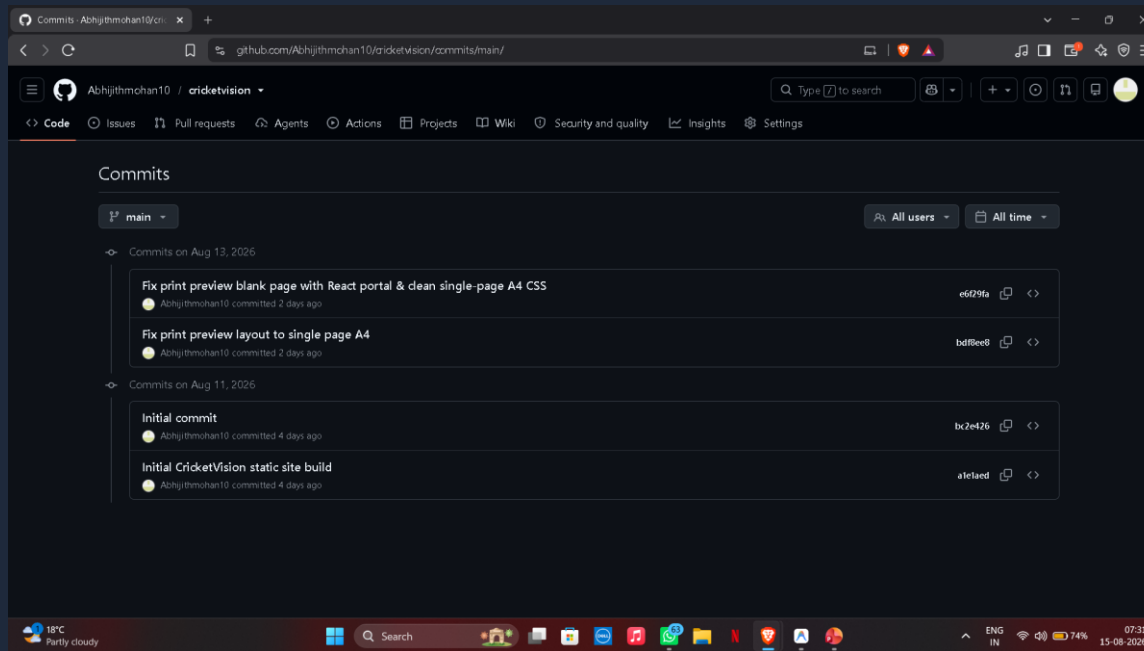
- **Successful MVP Delivery:** CricketVision successfully demonstrates a high-performance, full-stack cricket performance analytics platform.
- **Bridging Data & Coaching:** Empowers non-technical coaches and athletes with actionable visual insights, fatigue alerts, and AI coaching guidance.
- **High Academic & Commercial Potential:** Combines modern web technologies (React 18, Express, MongoDB) with innovative sports science algorithms.

Future Scope & Horizons

- **Live Score API Integration:** Connecting WebSocket feeds for live ball-by-ball match win predictions during international fixtures.
- **Deep Learning CV Engine:** Deploying custom MediaPipe pose models on GPU servers for frame-accurate 3D bowling action reconstruction.
- **Smart Wearable IoT Sync:** Integrating heart-rate and accelerometer smartwatch data for direct physiological fatigue tracking.

Git History & Version Control Log

Repository Commit Structure & Sprint Tracing



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